

Experiment - 8.1.2. Prime Numbers in a Given Range

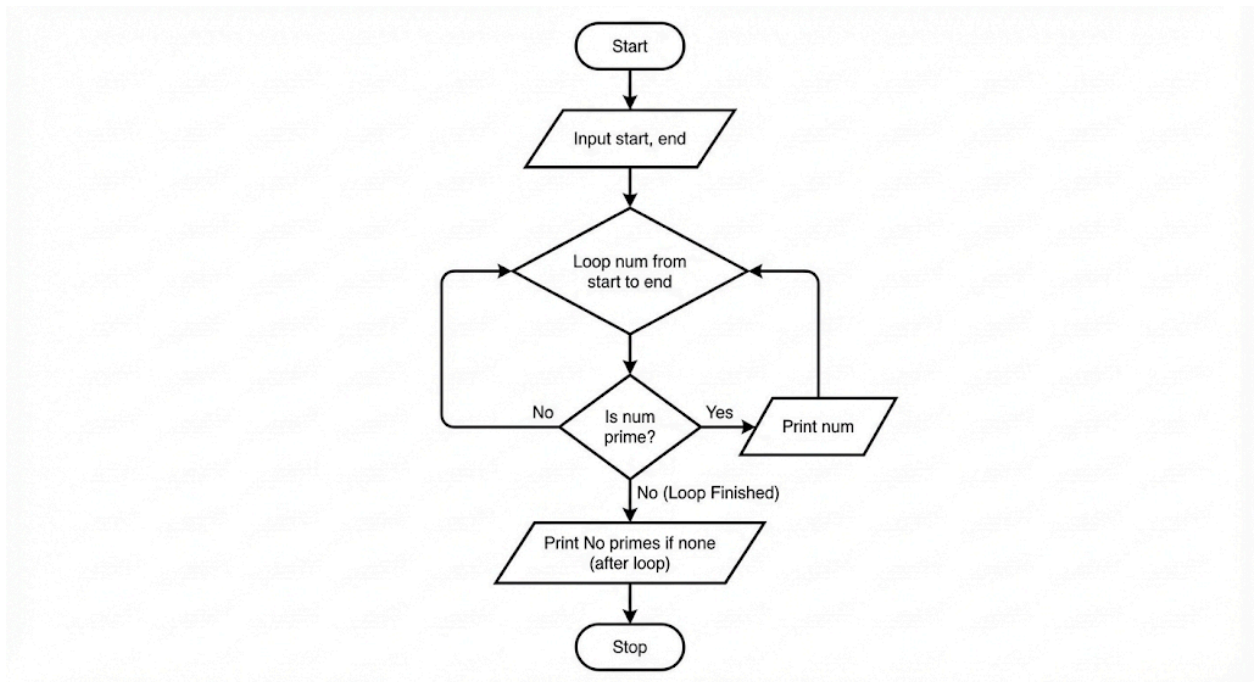
1. Aim

To design and implement a Python program that finds and prints all prime numbers within a specified range [start, end] (inclusive). The program determines the primality of each number and prints it. If no prime numbers exist within the given range, the program prints "No primes".

2. Pseudocode

1. **START**
2. **READ** an integer representing the start of the range and **STORE** it in start.
3. **READ** an integer representing the end of the range and **STORE** it in end.
4. **INITIALIZE** a boolean variable found to False to keep track of whether any primes exist.
5. **FOR** each number num in the range from start to end (inclusive):
 - **IF** num > 1 (since 1 and numbers below it are not prime):
 - **INITIALIZE** is_prime to True.
 - **FOR** each number i from 2 up to the integer square root of num ($\text{int}(\text{num}^{**0.5}) + 1$):
 - **IF** num is divisible by i ($\text{num} \% i == 0$):
 - **SET** is_prime to False.
 - **BREAK** the inner loop.
 - **IF** is_prime is True:
 - **PRINT** the prime number num.
 - **SET** found to True.
6. **IF** found is False (meaning no primes were printed):
 - **PRINT** "No primes".
7. **END**

3. Flowchart



4. Python Program

```
start = int(input())
end = int(input())
```

```
found = False
```

```
for num in range(start, end + 1):
    if num > 1:
        is_prime = True
        for i in range(2, int(num**0.5) + 1):
            if num % i == 0:
                is_prime = False
                break

        if is_prime:
            print(num)
            found = True
```

```
if not found:
    print("No primes")
```

5. Experiment Screenshot

8.1.2. Prime Numbers in a Given Range

Write a Python program to print all prime numbers within a given range (inclusive of both start and end values).

Input Format:

- First line contains an integer representing the start of the range.
- Second line contains an integer representing the end of the range.

Output Format:

- Print all prime numbers in the given range (including start and end if they are prime). Print each prime number on a separate line.
- If no prime numbers exist in the range, print: "No primes".

Sample Test Cases

```
1 start = int(input())
2 end = int(input())
3
4 found = False
5
6 for num in range(start, end + 1):
7     if num > 1:
8         is_prime = True
9         for i in range(2, int(num**0.5) + 1):
10            if num % i == 0:
11                is_prime = False
12            break
13         if is_prime:
14             print(num)
15         found = True
16
```

Average time: 0.004 s, Maximum time: 0.005 s

5 out of 5 shown test case(s) passed
4 out of 4 hidden test case(s) passed

Test case 1

Expected output

13
23
11
13
17
19

Terminal Test cases

Prev Reset Submit Next